

Aakash Kalmady

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EDUCATION

University of Maryland, College Park

College Park, MD

Bachelor of Science in Computer Science

Expected Graduation: May 2027

- GPA: 3.74/4.00, 3x Dean's List
- Relevant Coursework: Data Structures and Algorithms, Object-Oriented Programming, Computer Systems, Data Science, Machine Learning, Artificial Intelligence, Functional Programming, Discrete Math, Linear Algebra, Statistics and Probability

SKILLS

Languages: Python, Java, JavaScript, TypeScript, SQL (PostgreSQL), C, C++, Rust, OCaml, HTML, CSS, x86-64 Assembly

Frameworks: React.js, Next.js, Node.js, Express.js, Jest, Flask, Tailwind CSS

Developer Tools: AWS (S3, EC2), Google Firebase, Clerk, Neon, k6, Vim, Unix, Git, Valgrind, GDB, AutoCAD

Libraries: TensorFlow, OpenCV, Pandas, NumPy, Scikit-learn, Matplotlib, D3, Zod, Drizzle ORM

EXPERIENCE

Software Engineer

Sep 2025 - Present

Winrock International (via Hack4Impact-UMD)

College Park, MD

- Automating Winrock's client and supplier communications by developing a secure web app using React and Firebase
- Partnering with non-profit stakeholders in an Agile consulting team to adapt solutions in weekly sprints with a Kanban board
- Enabling secure auth and dynamic form handling to streamline client submissions and reduce manual coordination efforts

Software Engineer

Sep 2024 - Dec 2024

Booz Allen Hamilton (via App Development Club at UMD)

College Park, MD

- Automated Medicaid and CHIP claim auditing, reducing 12 hours of manual work per week and saving \$50,000 annually
- Improved user experience and accessibility by developing a dynamic front-end using Next.js, TypeScript, and TailwindCSS
- Collaborated with Booz Allen stakeholders in weekly sprints to align deliverables with the software development lifecycle
- Presented the product to policy specialist clients, effectively communicating software functionality and design decisions

Undergraduate Research Assistant

Jan 2024 - Dec 2024

University of Maryland, FIRE Program, Dr. Sara Strey

College Park, MD

- Researched and presented how sea surface temperature affects hurricanes by simulating Hurricane Ida under 3 conditions
- Built Python and shell scripts to automate the processing of over 100 GB of climate data from 3,000+ files on a remote, Linux-based Derecho HPC supercomputer
- Examined hurricane intensity patterns by documenting 15 hurricane data plots using wrf-python and shell scripts

PROJECTS

ClarityVue (clarityvue.com): Next.js, TypeScript, PostgreSQL, AWS S3, OAuth 2.0, Zod, Jest

Jun 2025 - Jul 2025

- Engineered a full-stack photo portfolio tool with authentication, album management, image uploads, and profile sharing
- Designed a scalable cloud infrastructure and back-end supporting over 1 TB of media and 500+ concurrent users using Amazon Web Services (AWS S3) for media storage and Neon serverless Postgres for high availability
- Reduced load times to under 200ms in production by leveraging Next.js SSR and optimizing complex database queries
- Protected against SQL injection attacks and enforced user-specific data access using OAuth 2.0 and Zod schema validation, leveraging Jest to test 20+ server actions, webhooks, and REST API endpoints for reliability

Eye Disease Detector: Python, Jupyter Notebook, Flask, AWS EC2, TensorFlow, OpenCV, Docker

Jul 2025 - Present

- Developing a machine learning model to classify three eye diseases by analyzing a dataset of over 4,000 retinal images
- Fine-tuning an ML model using OpenCV and NumPy for image processing and TensorFlow for a convolutional neural network (CNN) implementation, achieving a mean accuracy of 87.8% through 10-fold cross-validation
- Developing a REST API using Flask to integrate the ML model with an app to extract user image data for classification
- Deploying the application using Docker and designing it for scalable deployment on cloud infrastructure with AWS EC2

VEX Robotics: C++, V5 PROS API

Sep 2021 - May 2023

- Earned the design award at the world championship, competing with 800 teams, and the title of national champions
- Developed C++ programs and feedback controllers using 10 different sensors for the autonomous navigation of a robot
- Fine-tuned controllers by finding optimal settings in Excel for 50% more precision within a 1% error margin (± 0.2 inches)
- Documented results in a 500+ page notebook to illustrate the design process and project management of our team